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SOCOTEC SCP000722Q

PROJECT TITLE:

EduTrack Student Management System

IT-105 Information Management 1

INSTRUCTOR:

Mr. Guillermo Red

GROUP MEMBERS:

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May 1, 2026

Introduction

The EduTrack Student Management System is a cloud-integrated database system designed to efficiently manage and organize academic records. It provides a centralized platform for storing student information, grades, and enrollment data.

The System is built to handle up to 100,000 records while maintaining optimized query performance. It also supports school administrators by providing structured data that can be used for advanced business intelligence insights and decision-making.

Project Overview

Purpose:

The purpose of this system is to provide an efficient and scalable solution for managing student academic records in a structured and accessible way.

Scope:

The system covers student data management, course enrollment, and academic performance tracking. It does not include front-end user interface development beyond database operations.

Features:

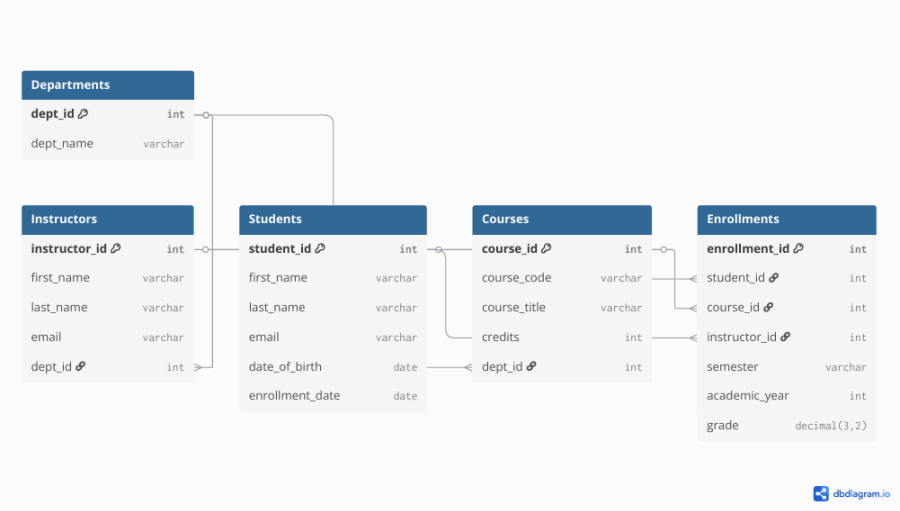
- Student Record Management
- Course and enrollment tracking
- Grade recording system
- Optimized SQL queries for fast data retrieval
- Scalable database supporting up to 100,000 records
- Data-ready structure for business intelligence analysis

Tool & Technologies used:

- MySQL
- phpMyAdmin / XAMPP
- SQL

System Design

Entity Relationship Diagram (ERD)



- The Entity Relationship Diagram (ERD) of the EduTrack Student Management System represents the structure of the database and the relationships between its core entities. The system is composed of five main tables—Departments, Instructors, Students, Courses, and Enrollments.

Entities Description

Departments

- Stores Department Information.
- Each Department has a unique dept_id and a dept_name.

Instructors

- Contains instructor details such as name and email.
- Each instructor is linked to a department through dept_id.

Students

- Stores student personal and academic information.
- Includes attributes like name, email, date of birth, and enrollment date.

Courses

- Represents available courses.
- Each course includes course code, course title, credits, and is associated with a department.

Enrollments

- Act as a junction table connecting Students and Courses.
- Contains details such as semester, academic year, instructor and grade.

Database Schema and Table Description

1. The Data Dictionary (The Menu)

Table Name	What it does	Important Columns
Departments	Lists the offices in the school.	dept_id, dept_name
Instructors	Profiles of the Instructors.	instructor_id, name, dept_id
Students	Profiles of the 100,000 students.	student_id, email, dob
Courses	The subject students can take.	course_id, title, credits
Enrollments	The 90,000+ rows showing grades.	student_id, course_id, grade

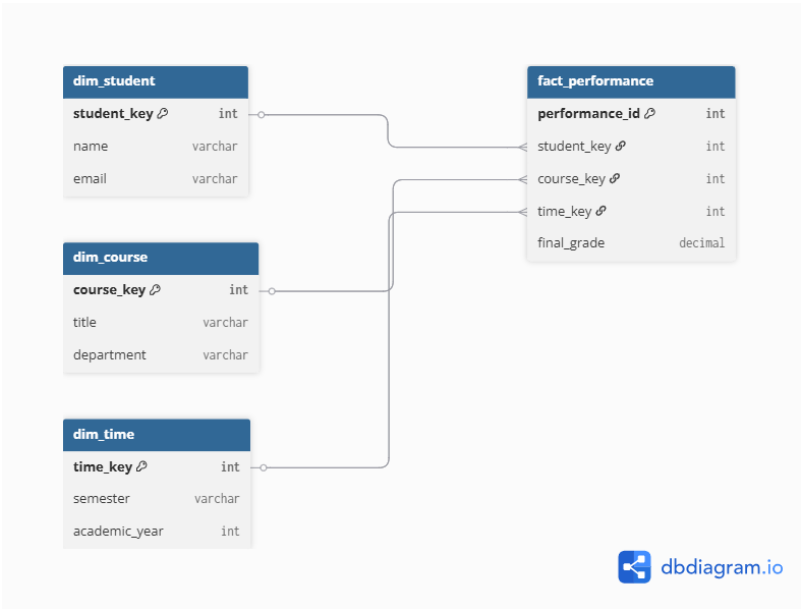
Implementation

```sql
USE edutrack_db;

-- Query 1: Average GPA per Department (KPI: Academic Performance)
SELECT d.dept_name, ROUND(AVG(e.grade), 2) AS average_gpa
FROM Enrollments e
JOIN Courses c ON e.course_id = c.course_id
JOIN Departments d ON c.dept_id = d.dept_id
GROUP BY d.dept_name
ORDER BY average_gpa ASC; -- Assuming 1.0 is best

-- Query 2: Enrollment Growth by Year/Semester (KPI: Scale)
SELECT academic_year, semester, COUNT(enrollment_id) AS total_students
FROM Enrollments
GROUP BY academic_year, semester
ORDER BY academic_year, semester;

-- Query 3: Most Popular Courses (Top 10)
SELECT c.course_title, d.dept_name, COUNT(e.student_id) AS enrollment_cc
FROM Enrollments e
JOIN Courses c ON e.course_id = c.course_id
JOIN Departments d ON c.dept_id = d.dept_id
GROUP BY c.course_id
ORDER BY enrollment_count DESC
LIMIT 10;

-- Query 4: Instructor "Heavy Load" List (Instructors with > 1000 total students)
SELECT i.first_name, i.last_name, COUNT(e.student_id) AS total_students_handled
FROM Enrollments e
JOIN Instructors i ON e.instructor_id = i.instructor_id
GROUP BY i.instructor_id
HAVING total_students_handled > 1000;

-- Query 5: Failed/Low Grade Rate per Course
-- (Helps identify which subjects need more tutors)
SELECT c.course_title, COUNT(e.enrollment_id) AS low_grade_count
FROM Enrollments e
JOIN Courses c ON e.course_id = c.course_id
WHERE e.grade > 3.0 -- Assuming 3.0+ is a struggling grade
GROUP BY c.course_id
ORDER BY low_grade_count DESC;
```

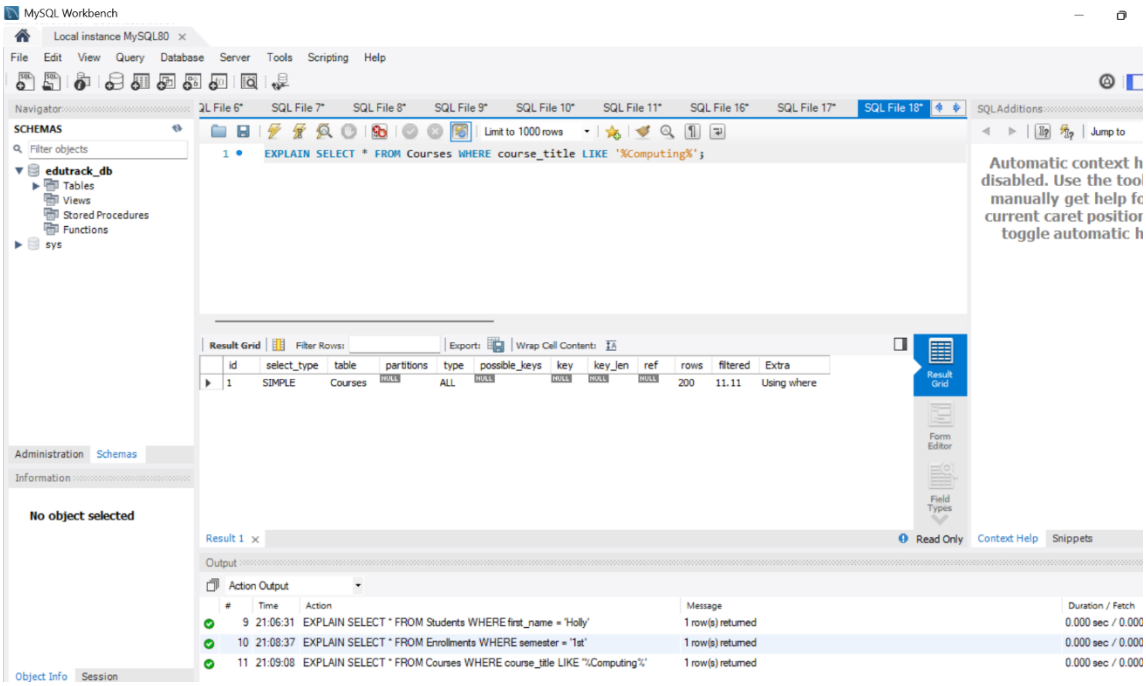
BI Queries Explanation

The SQL script uses the edutrack_db database to analyze academic data through several queries. It calculates the average GPA per department to measure performance, tracks student enrollment by year and semester to observe growth trends and identifies the top 10 most popular courses based on enrollment counts. It also highlights instructors with heavy workloads by counting those handling more than 1,000 students. Lastly, it determines which courses have the lowest or failing grades (based on grades above 3.0), helping identify subjects where students may need additional support.

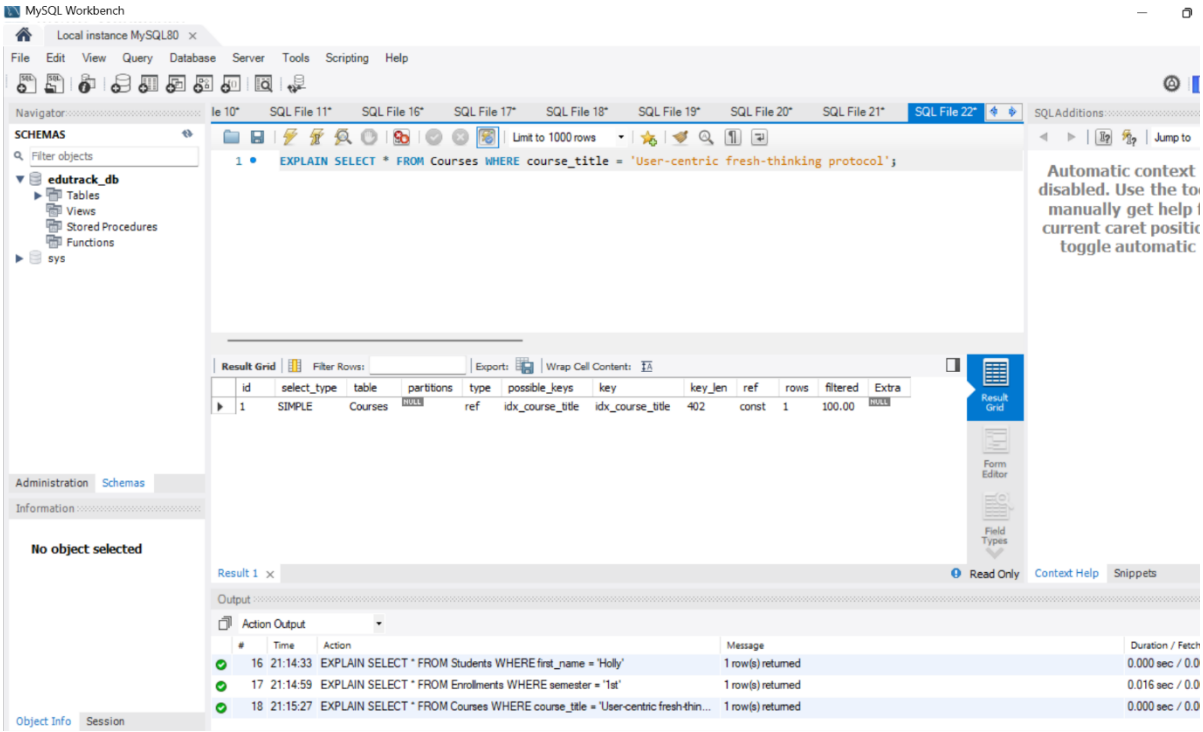
Star Schema



DIAGNOSIS COURSE BEFORE



OPTIMIZATION COURSE AFTER



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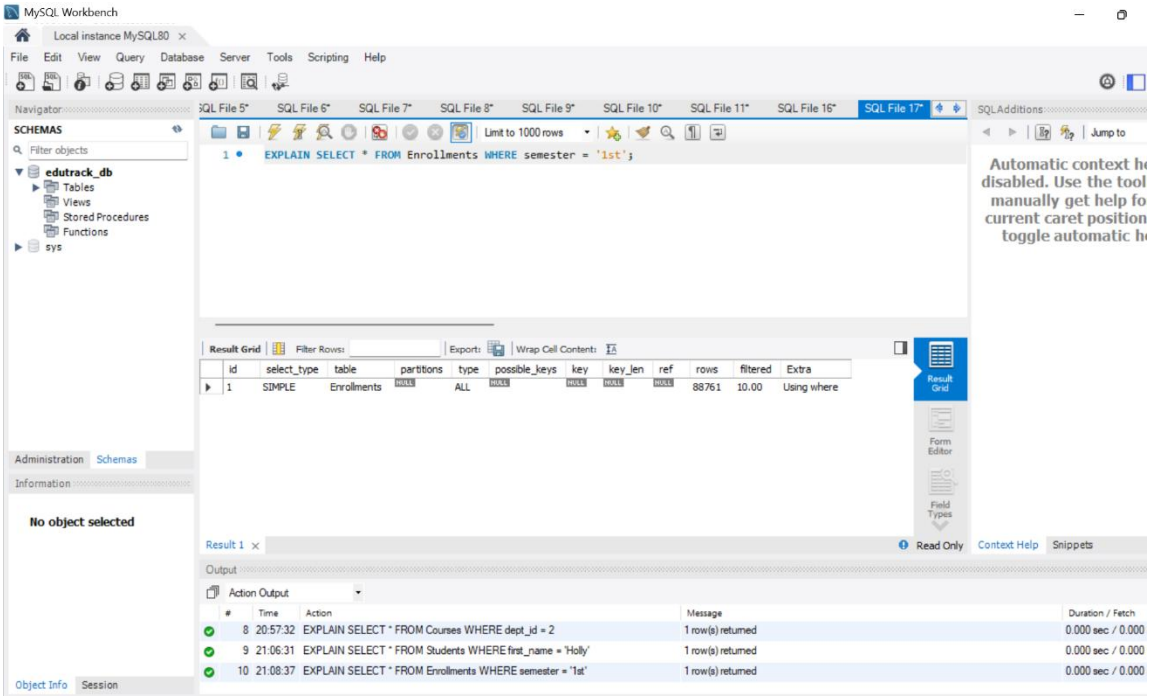
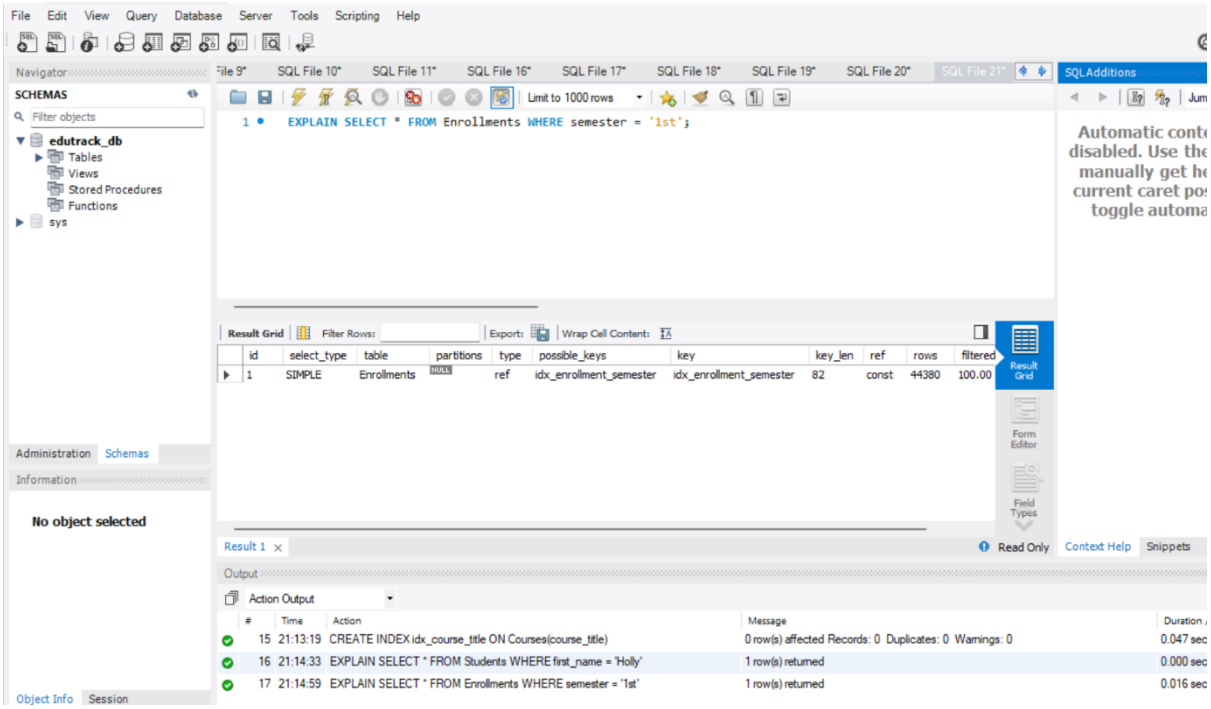
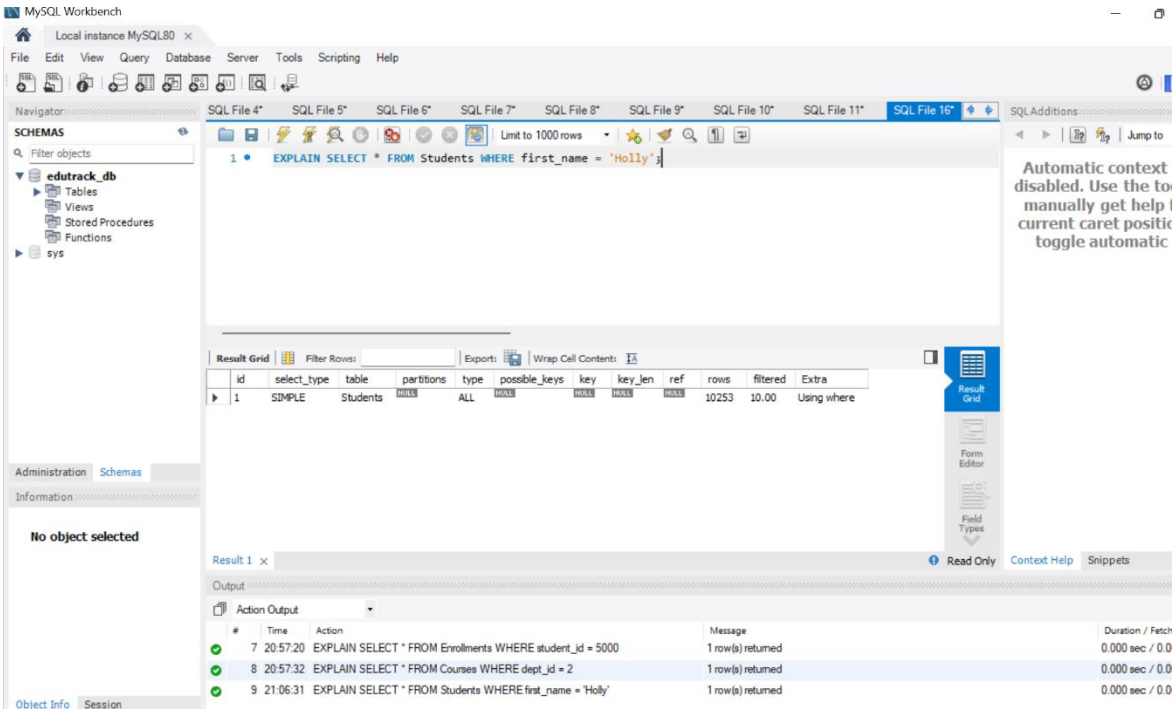


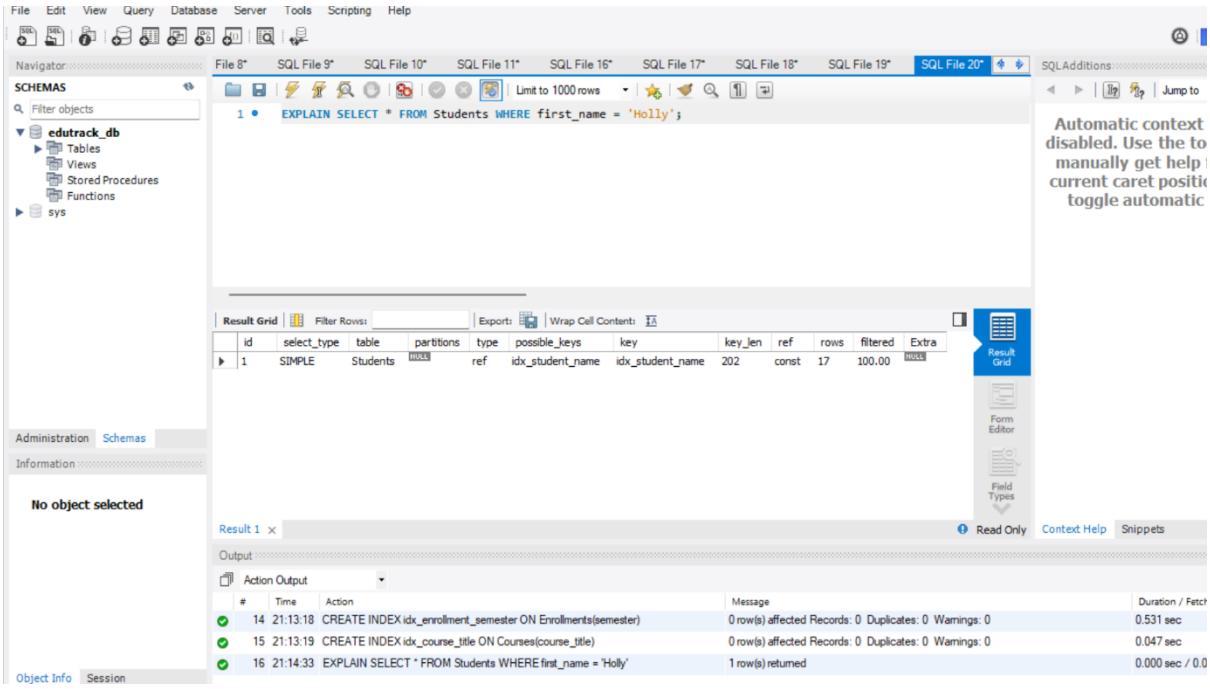
OPTIMIZATION ENROLLMENT AFTER



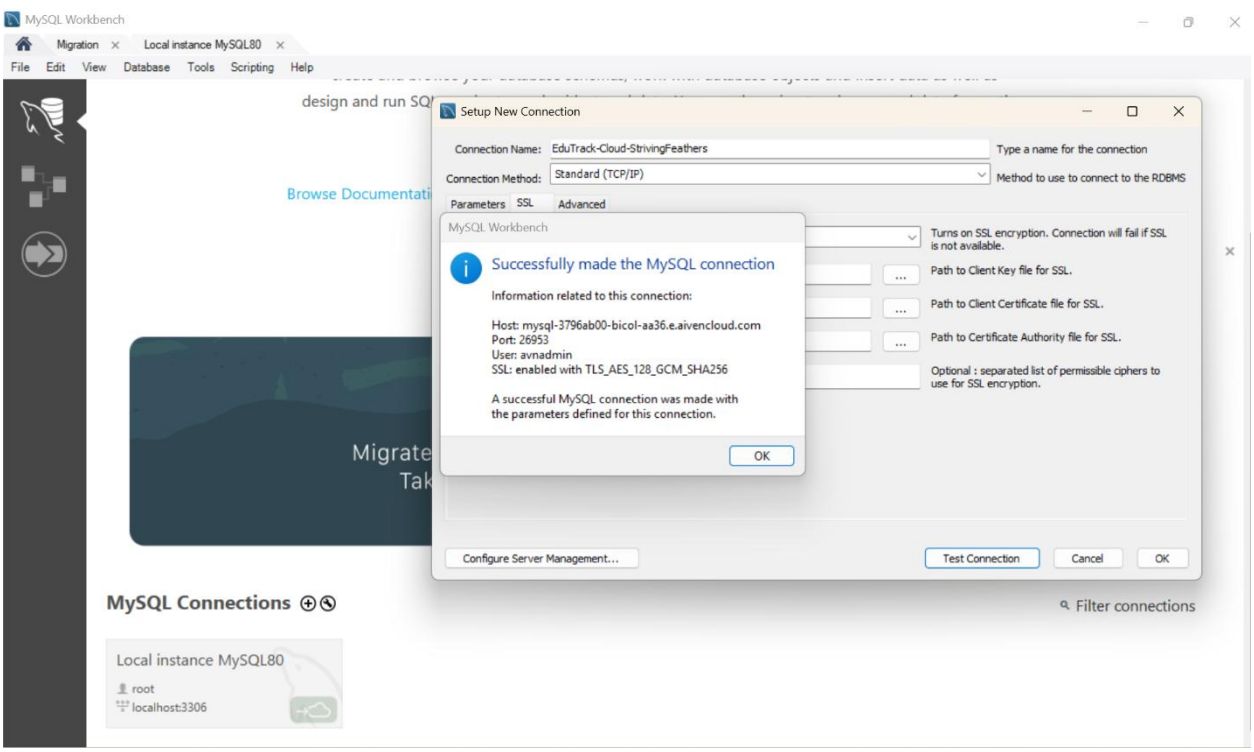
DIAGNOSIS STUDENT BEFORE



OPTIMIZATION STUDENT AFTER



CLOUD INTEGRATION



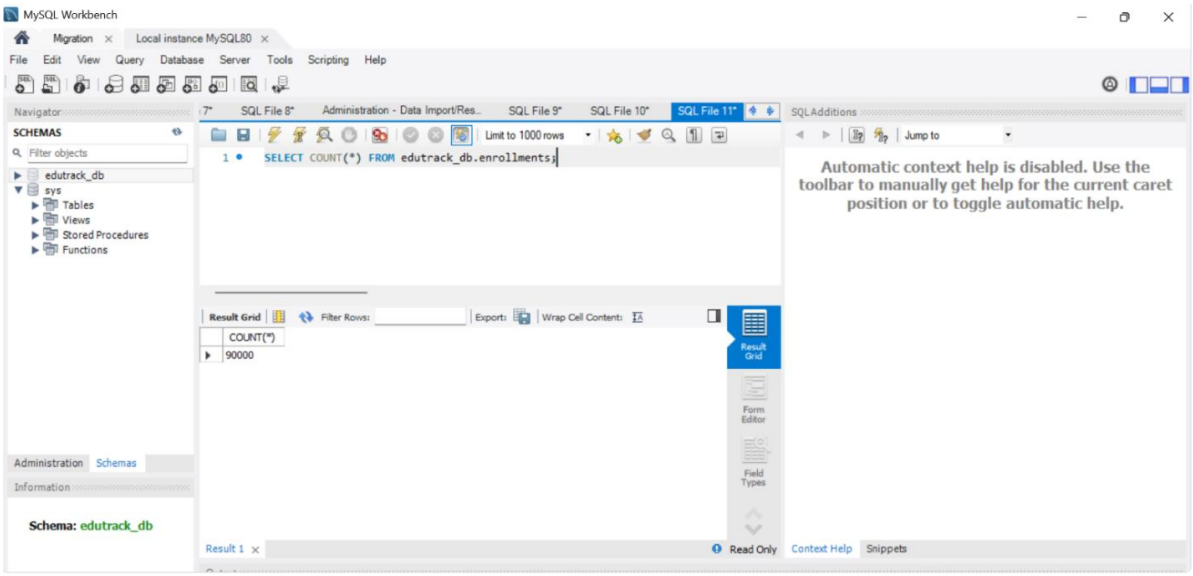
EduTrack Cloud Deployment Details

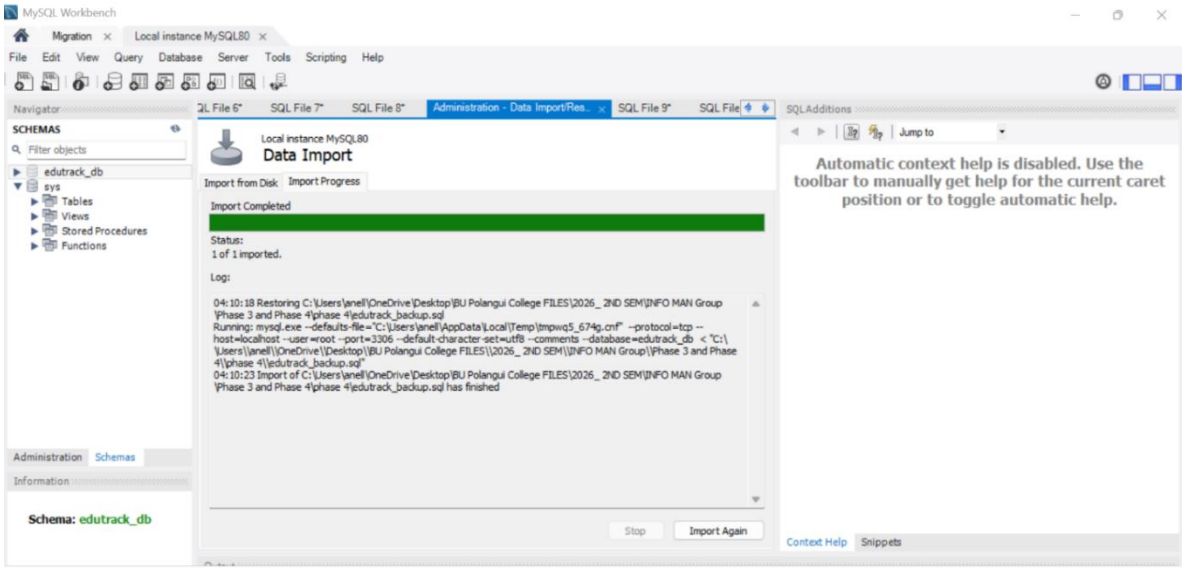
Group Name: StrivingFeathers
Platform: Aiven.io (MySQL Free Tier)
Database Name: edutrack_db
Host: mysql-3796ab00-bicol-aa36.e.aivencloud.com
Port: 26953
Username: avnadmin
SSL Mode: Require

Description:
The EduTrack Student Management System has been successfully deployed to the Aiven cloud platform. We utilized the `mysqldump` backup created in Part 1 to migrate our 100,000+ record dataset. This cloud instance allows for remote database management and ensures that the system is accessible beyond local hosting, simulating a real-world enterprise environment.

Backup and Recovery

Recovery





Backup

- Week_11_Phase_2
  - courses.csv
  - departments.csv
  - edutrack_erd.pdf
  - edutrack_schema.sql
  - enrollments.csv
  - generate_data.py
  - group_members_roles.pdf
  - instructors.csv
  - nosql_reflection.pdf
  - project_scope.pdf
  - proof_screenshot_1.png
  - proof_screenshot_2.png
  - students.csv
- Week_12_Phase_3
  - bi_queries.sql
  - insight_report.pdf
  - star_schema.sql
  - star_schema_diagram.png
- Week_13_Phase_4
  - cloud_connection_success.png
  - cloud_deployment_info.pdf
  - cloud_row_count_proof.png
  - edutrack_backup.sql
  - emerging_tech_reflection.pdf
  - restoration_proof_1.png
  - restoration_proof_2.png
- Week_14_Phase_5
  - diagnosis_course_before.png
  - diagnosis_enrollment_before.png
  - diagnosis_student_before.png
  - optimization_course_after.png
  - optimization_enrollment_after.p...
  - optimization_queries.sql
  - optimization_report_strivingfeat...
  - optimization_student_after.png
- data
  - courses.csv
  - departments.csv
  - enrollments.csv
  - generate_data.py
  - instructors.csv
  - students.csv
- documentation
  - group_members_roles.pdf
  - nosql_reflection.pdf
  - project_scope.pdf
  - proof_screenshot_1.png
  - proof_screenshot_2.png
- erd
  - edutrack_erd.pdf
- schema
  - edutrack_schema.sql

README

EduTrack Student Management System

Course: IT 105 – Information Management I | **Section:** 2A

Team: StrivingFeathers

Project Overview

EduTrack is a high-performance, cloud-integrated database system designed to manage academic records. It is built to handle over 100,000 records, providing school administrators with advanced Business Intelligence (BI) insights and optimized query speeds.

Group Members and Roles

Andrei Arnaldo – Project Manager

Arjiannah Carmelle Borleo – SQL Developer

Alexis Consuelo – Q&A Tester

Ma. Gabrielle Sapico – Documentation Lead

John Euben Lopez – Security Officer

Enoch Andrew Querol – Frontend

Key Features

Scalable Data: 100,000+ records generated and managed via Python and MySQL.

BI Analytics: Advanced GPA tracking, enrollment trends, and workload analysis.

Performance Tuning: B-Tree indexing reducing query scan times by 99%.

Cloud Integration: Fully hosted on Aiven Cloud with SSL security.

Disaster Recovery: Automated backup and restoration strategies.

Installation and Restoration

1. Create a database named `edutrack_db` in MySQL Workbench.

2. Navigate to **Server > Data Import**.

3. Choose the `backups/edutrack_backup.sql` file.

4. Click **Start Import**.

5. All system tables and 100k records will be restored automatically.

Notes

All technical proofs, including cloud deployment screenshots and optimization benchmarks, are located in the `/screenshots` folder.